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The impact of population aging on corporate digital transformation: Evidence from China

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ABSTRACT

Labor shortages caused by population aging may directly compel enterprises to undergo digital transformation or indirectly pose a continuous challenge owing to insufficient talent reserves. Currently, no studies have explicitly explored the impact of population aging on corporate digital transformation. To clarify this relationship, this study integrates population aging and corporate digital transformation into the same theoretical framework and uses panel data from China's Shanghai and Shenzhen A-share listed companies. A multiple linear regression model is employed for theoretical analysis and empirical testing. The findings indicate that, first, population aging promotes corporate digital transformation, a conclusion that remains robust after a series of robustness checks and consideration of endogeneity issues. Second, the impact of population aging on corporate digital transformation exhibits significant heterogeneity across industries and regions. Specifically, the aging population has a more pronounced effect on the digital transformation of labor-intensive enterprises and companies located in the eastern regions, where the degree of aging is more pronounced. The conclusions of this study provide empirical evidence and policy implications for actively responding to population aging and advancing corporate digital transformation in the context of an aging population.

1. Introduction

With the rapid development of emerging information technologies, such as artificial intelligence, blockchain, cloud computing, and big data, digital technology has become the core force driving a new round of global technological revolution and industrial transformation. These technologies are regarded as disruptive innovations that "change the rules of the game" and "reshape the future landscape." Countries worldwide are strategically planning and proactively positioning themselves to accelerate the application of digital technologies in enterprises, aiming to seize the benefits of future technological and industrial development. However, digital transformation is a complex and challenging task that imposes high demand on corporate talent reserves. Population aging is an important factor affecting the labor supply and human capital reserves (Topel, 1999; Jing and Li, 2024). Therefore, the intensification of population aging will inevitably affect corporate digital transformation and may pose continuous and profound challenges.

Population aging has become a nearly universal trend in global population development.[1] Unfortunately, academic literature explicitly exploring the impact of population aging on corporate digital transformation is scarce. Given that digital technology can fundamentally change consumer behavior and expectations and disrupt the competitive landscape of incumbents (Vial, 2021), it is considered a disruptive innovation. Innovation theory provides inspiration for this study. Existing research shows that technological innovation is influenced by factors such as religious culture, democratization, global leadership, the technical division of labor, and resource cooperation between organizations (Coccia, 2010, 2014a, 2014b, 2015a, 2015b). Technological innovation is also closely related to a country's demographic factors, such as population growth rate and structure, especially population

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[1] The United Nations recently published the "World Population Prospects: The 2024 Revision – Key Findings," which projects that by the end of the 2070s, the global population aged 65 and older will surpass the number of individuals under the age of 18.

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aging (Coccia, 2014c; Ang and Madsen, 2015). However, no consensus has been reached in the related literature. Aging is accompanied by "cognitive aging," including declines in reasoning ability, spatial orientation, numerical ability, memory, problem-solving skills (Engelhardt et al., 2010), innovative consciousness (Kanfer and Ackerman, 2000), and innovative capacity (Czaja and Lee, 2007), which may negatively affect technological innovation. Furthermore, population aging leads to labor shortages and increased labor costs, changing the relative cost and structure of labor and technological elements, which may compel organizations to implement technological innovations (Acemoglu and Restrepo, 2022; Irmen, 2009, 2017).

Unlike technological innovation, corporate digital transformation is the behavioral application of digital technology that further drives customer experience optimization and creates new business operation models through comprehensive organizational changes (Verhoef et al., 2021). The differences between digital transformation and technological innovation may lead to different effects of population aging. Therefore, further in-depth theoretical and empirical analyses are required to explore this topic.

Accordingly, our study uses panel data from Chinese listed companies to analyze and discuss the following questions: (1) Does population aging have a substantial impact on corporate digital transformation? (2) Does the impact of population aging on corporate digital transformation show heterogeneity? China's practices are more representative in addressing these issues. First, China is the largest emerging market country in the world and one of the fastest aging countries.[2] Since implementing the reform and opening-up policy in 1978, China has rapidly integrated into the global value chain division system by leveraging its low labor cost advantage and promoting rapid national economic development. However, population aging makes this mode of economic growth unsustainable, indicating the urgent need for a transition from factor-to technology-driven economic development. Thus, an active response to rapid population aging has become a national strategy in China (Peng et al., 2023). Second, in recent years, Chinese enterprises have developed many best practices for the application of digital technologies worldwide, such as the integrated application WeChat, payment application Alipay, and social entertainment application Douyin, providing a rich research foundation for Chinese scholars (Chen and Tian, 2022). Our findings show that population aging promotes corporate digital transformation, which remains valid even after considering a series of robustness tests and endogeneity issues. The heterogeneity test reveals that the impact of population aging on corporate digital transformation is heterogeneous, with a greater promotion effect observed for labor-intensive enterprises and those in the eastern region.

The remainder of this paper is organized as follows: Section 2 provides a literature review; Section 3 presents the theoretical analysis and research hypotheses; Section 4 describes the research design, including samples, data, statistical models, and variable explanations; Section 5 provides the empirical analysis and robustness tests; Section 6 covers the heterogeneity tests; and Section 7 summarizes the conclusions, implications, and limitations.

2. Literature review

2.1. Research on population aging

According to the "Dictionary of Demography" compiled by the International Union for the Scientific Study of Population, population aging refers to a society in which the proportion of the population aged 60 and above reaches or exceeds 10 % of the total, or the proportion of those aged 65 and above reaches or exceeds 7 %, termed as an "aged population" society. Scholars believe that population aging primarily affects factor and consumer markets.

The impact of population aging on production factors is reflected mainly in changes in the labor market. First, population aging directly leads to a shortage of labor resources and rising labor costs, altering the relative costs and structure of technological and labor elements in the factor market (Acemoglu and Restrepo, 2022; Irmen, 2009, 2017). Second, owing to differences in labor productivity among different age groups, increased population aging negatively affects overall labor productivity (McMillan and Baesel, 1990; Leibfritz and Roeger, 2008; Cai and Stoyanov, 2016).

In consumer markets, the age structure of the population significantly affects consumption (Modigliani and Brumberg, 1954). The unique consumption needs, habits, preferences, and abilities of older adults greatly influence their social consumption structures (Aigner-Walder and Döring, 2012). This manifests in several aspects. First, the needs of the aging population are more personalized. Owing to physical conditions (e.g., mobility issues, declining vision and hearing, and chronic diseases), lifestyle habits, and hobbies, older adults have varied pain points and personalized needs, requiring customized products and services, such as personalized health management plans and customized home facilities. Second, the needs of the aging population are becoming increasingly fragmented. The products and services required during the aging process change and are gradually being replaced, thereby exhibiting fragmented characteristics. Third, the aging population requires convenient and intelligent products and services. As individuals age, their physiological functions gradually deteriorate, and their movements become less agile, leading to a preference for convenience and intelligence in consumption.

New business models based on digital technology must be developed to meet these personalized, fragmented, convenient, and intelligent consumption needs and enable older adults to live independent and healthy lives (Oderanti et al., 2021). Over the past decade, digital technology has advanced rapidly, becoming an important tool for addressing the effects and challenges of population aging. For example, Moreno-Sánchez et al. (2024) explored how the health and care needs of older adults change with age and how digital technology can be used to meet these personalized needs. Canali et al. (2024) noted that the decline in physical function and presence of chronic diseases lead to decreased face-to-face interactions among older adults, whereas digital technology can prevent, detect, and manage health conditions to meet the needs of the aging population. Therefore, the application of digital technology in healthcare has rapidly increased in recent years. A study based on Japan's aging society also found that wearable digital technology devices can provide remote medical treatment and health monitoring, which is beneficial for older adults living at home who experience chronic diseases and other illnesses (Tanaka et al., 2024). However, digital technology also raises ethical issues, such as user privacy, security, and autonomy, causing concerns among older adults (Campbell et al., 2024).

In summary, population aging directly affects factor markets by reducing the supply of labor elements. It also affects consumer markets through changes in consumption demand, further challenging traditional business models. Digital technology has become one of the most effective means of mitigating the impact of population aging. In socioeconomic activities, enterprises are both demanders of various production factors, purchasing elements to produce goods and services, and providers of goods and services in the consumer market. Therefore, changes in the factor and consumer markets caused by population aging will inevitably affect corporate decision-making. However, few studies have explored the impact of population aging on corporate decisions from a micro-enterprise perspective, particularly in the context of digital transformation. Thus, the present study addresses this research gap.

[2] According to official Chinese statistics, as of the end of 2023, China's population has experienced negative growth for two consecutive years, with individuals aged 65 and older constituting 15.4 % of the total population.

2.2. Research on digital transformation

The academic community has extensively defined the concept of digital transformation. According to Vial (2021), from a process perspective, digital transformation is a mechanism that triggers significant changes in entities through the combined application of information, computing, communication, and connectivity technologies. Several scholars have focused on the outcomes of transformation, emphasizing the impact of digital transformation on business model innovation. For example, Verhoef et al. (2021) defined digital transformation as the corporate utilization of digital technologies to develop new business models, thereby aiding businesses in creating and distributing value. Although existing studies have not reached a consensus on the definition of digital transformation, it is evident that it is characterized by the (1) application of digital technology and (2) integration of digital technology that alters business models (Ciriello et al., 2018).

A business model refers to the mechanisms through which an organization connects with external stakeholders, thereby creating value for customers and partners through collaborative and managed economic activities (Teece, 2010). Traditional business models focus on physical assets and linear processes, typically involving fixed production and sales procedures and direct interactions with consumers through physical channels to capture value. In these models, corporate success often depends on investments in and management of physical assets, which lack flexibility and the ability to adapt quickly to market changes. The openness, distribution, editability, re-combinability, accessibility, and generative nature of digital technologies (Kallinikos et al., 2013) have altered how enterprises create and capture value (Yoo et al., 2010, 2012). New business models leverage digital technologies to achieve operational flexibility and scalability, enabling companies to respond rapidly to market changes and provide personalized services to meet evolving consumer demands. This is primarily reflected in the following aspects. First, digital technologies enable companies to optimize business processes quickly and in a timely manner, thereby reducing production costs and improving efficiency (Li, 2020). Second, digital technologies facilitate the perception and capture of more heterogeneous resources from consumers, allowing companies to more accurately match changing demand preferences and reshape their methods of creating, delivering, and capturing value (Autio et al., 2018). Third, they enable companies to coordinate geographically dispersed audiences, thus focusing on a broader ecological perspective and opening new ways to build and utilize platform capabilities (Nambisan et al., 2017; Yoo et al., 2012) by promoting the grafting of resources among different entities to create and transfer value.

The driving factors of digital transformation (i.e., how to drive a company's digital transformation) have also received significant academic attention, with discussions mainly focusing on two aspects. First, the influence of individuals as decision-makers and implementers on corporate digital transformation has been explored. Through surveys of 938 Portuguese companies from different countries, Ferreira et al. (2019) found that younger, female, and highly educated managers were more supportive of corporate digital innovation. AlNuaimi et al. (2022) discovered that managers with a digital transformation mindset positively influence an organization's digital transformation. Second, the impact of organizational capability factors has been examined. Dremel et al. (2017) suggested that cross-departmental collaboration is a condition for conducting big data analysis and undertaking digital transformation. Warner and Wäger (2019) posited that during the digital transformation process, companies must establish strong dynamic capabilities to quickly create, implement, and change business models. Using six Italian small and medium-sized enterprises as case studies, Matarazzo et al. (2021) confirmed the impact of corporate dynamic capabilities on digital transformation.

Although digital transformation results from the coevolution of enterprises with their internal and external environments (Chen and Tian, 2022), few studies have explored the role of the external environment in driving corporate digital transformation. Population aging is one of the most important external environments that enterprises must face long term. Thus, a systematic analysis and exploration of the impact of population aging on corporate digital transformation are needed, providing an opportunity for this study.

2.3. Research on population aging and technological innovation

To date, no study has comprehensively examined the effects of population aging on corporate digital transformation. However, insights from technological innovation theory offer a foundational understanding of this issue. The existing literature suggests that population aging significantly influences technological innovation, although a consensus among studies remains elusive.

Some scholars have argued that population aging positively affects technological innovation. An aging population leads to a decline in the labor supply and an increase in labor costs, altering the relative cost structure of technology versus labor. This shift incentivizes organizations to innovate technologically to replace labor inputs (Acemoglu and Restrepo, 2022; Irmen, 2009, 2017). For example, Abeliansky and Prettner (2017) analyzed panel data from 60 countries from 1993 to 2013 and found that nations with lower population growth rates had stronger incentives to invest in automation technologies as labor substitutes, thereby fostering the development of these technologies. Acemoglu and Restrepo (2022) highlighted how demographic changes, particularly aging, drive the development and application of robotics and other industrial automation technologies. They noted that the relatively slower adoption and application of automation technologies in the United States and the United Kingdom compared to Japan, South Korea, and Germany can be partially attributed to the former's lower degree of aging. From a human capital accumulation standpoint, older adults bring extensive work experience and theoretical knowledge accumulated over their careers, thereby enhancing their ability to contribute to the underlying logic of technological research and development (R&D). Their “learning-by-doing” approach enables better job matching and adaptation, which ultimately boosts innovation efficiency (Ang and Madsen, 2015). For example, Sheng et al. (2024) discovered a positive correlation between population aging and corporate green technology innovation.

Conversely, other researchers have focused on aspects such as human capital, innovation capacity, and motivation, arguing that population aging impedes technological innovation. From a human capital perspective, Verhaeghen and Salthouse (1997) found that aging diminishes cognitive abilities and reduces learning capacity in older adults. Meyer (2009) indicated that firms with a higher proportion of older employees struggle more with adopting new technologies than those with younger employees. As individuals age, they perceive limited future benefits from learning new skills, particularly when the advantages conferred by innovative technologies are delayed. Consequently, older adults may resist engaging in additional technology training or education (Friedberg, 2003). Additionally, population aging escalates labor costs for businesses and social security expenditures, intensifying corporate tax burdens. This financial strain could potentially “crowd out” investments in R&D, adversely affecting technological innovation (Behaghel and Greenan, 2010).

Moreover, some scholars have proposed a nonlinear relationship between population aging and technological innovation. For example, Coccia (2014c) used data from 34 OECD countries spanning 1985 to 2005 to demonstrate an inverted U-shaped association between population growth rates and technological innovation.

This literature review underscores the depth of investigations into population aging and its implications for technological innovation, providing valuable theoretical underpinnings for this study. Nevertheless, unlike general technological innovation, corporate digital transformation represents a transformative endeavor that leverages digital

technologies to forge novel business models. Hence, the specific effects of population aging on corporate digital transformation warrant further exploration.

3. Theoretical analysis and research hypotheses

This study explores the influence of population aging on corporate digital transformation through two primary mechanisms grounded in the interplay between demographic shifts and technological advancement.

First, an aging population precipitates a diminishing labor force, compelling firms to adopt digital transformation as a strategic imperative. Innovation often stems from problem-solving endeavors. Thus, when confronted with labor shortages and escalating labor costs, companies are incentivized to seek innovative solutions to maintain their competitive advantage and profitability (Coccia, 2017a, 2017b). The shifting economics of production factors, wherein digital technologies become more cost-effective relative to human labor, fuels the adoption of automation and AI as substitutes for manual tasks. Evidence suggests that digital technologies not only supplant labor but also enhance efficiency and reduce operational costs (Shibata, 2019). For example, AI facilitates automated decision-making and replaces roles characterized by low skill requirements, high repetition, and high substitutability, thereby reducing reliance on labor. Cloud-based "human cloud" models enable global talent sourcing and foster flexible project-based collaborations. Furthermore, digitally empowered smart workers and production lines operate tirelessly, transcending human limitations.

Second, evolving consumer preferences among an aging demographic necessitate digital transformation to cater to emerging societal demands. Older adult consumers exhibit distinct consumption patterns marked by personalization, convenience, and fragmentation, thereby challenging conventional business paradigms anchored in rigid production schedules and direct physical engagement with customers. To adapt, businesses must leverage digital technologies to cultivate novel business models that are attuned to an aging society (Oderanti et al., 2021). These models foster operational agility and scalability, enabling swifter responses to market fluctuations and tailored offerings that resonate with diverse consumer segments. Digital innovations disrupt traditional communication hierarchies and facilitate bidirectional interactions in which consumers co-create value. Data-driven processes streamline operations, mitigate delays, and enhance efficiency. Smart manufacturing systems seamlessly transition between batch sizes, accommodating individualized and customizable products. In complex scenarios that exceed a single firm's capabilities, digital transformation fosters inter-enterprise collaboration via networked ecosystems, thus amplifying collective value creation potential. The Centre for Strategic and International Studies (CSIS) has underscored the pivotal role of digital transformation in addressing the challenges posed by population aging (CSIS Report).

Therefore, we posit the following hypothesis:

H1a. Population aging promotes corporate digital transformation.

Population aging may negatively affect corporate digital transformation by reducing the human capital supply and constricting innovation resources. For enterprises, digital transformation is a novel arena filled with uncertainties and challenges. Thus, firms must develop dynamic capabilities and leadership while leveraging existing competencies (Warner and Wäger, 2019; AlNuaimi et al., 2022), and demand a substantial workforce of information technology specialists. This elevates the standards of both human capital accumulation and financial investment by enterprises. However, as population aging intensifies, the age profile of corporate labor shifts towards an older demographic. Compared with younger employees, older workers exhibit a diminished capacity to acquire new knowledge and adapt to novel technologies (Verhaeghen and Salthouse, 1997). The myopic focus on immediate benefits and heightened risk aversion among older employees discourages them from embracing new technological learning or training (Friedberg, 2003). Consequently, in the face of groundbreaking digital technologies and the potential fundamental overhaul of business operations, aging employees may adopt a conservative or even negative stance, showing reluctance to abandon familiar methodologies in favor of mastering and implementing new digital tools and processes. Thus, the often lack essential digital literacy and adaptability. This diminishes the availability of human resources during corporate digital transformation and impedes progress. Concurrently, an aging population contributes to escalating labor costs and social security outlays for companies, leading to augmented tax burdens that could exert a "crowding-out effect" on corporate R&D investments and adversely affect digital transformation endeavors.

Furthermore, population aging may undermine corporate digital transformation by altering consumption patterns. Older adults generally have more conservative consumption tendencies, with a predilection for conventional services and products and a lower receptivity to emerging digital offerings. Empirical studies indicate that older individuals encounter multiple barriers in acquiring and utilizing digital technology (Li and Kostka, 2024; Černá et al., 2023), and older adult investors demonstrate reduced demand for digital inclusive finance (Zhou and Ye, 2023). Such consumption behavior can result in inadequate market demand when corporations embark on digital transformation projects. If an enterprise's primary customer base comprises the aging population, it must contemplate strategies to harmonize online and offline services effectively and enhance the uptake of digital products among older adults through targeted user education and marketing initiatives.

In light of these considerations, we posit the following counter hypothesis:

H1b. Population aging hinders corporate digital transformation.

4. Research design

4.1. Sample and data

The CSMAR, CNRDS, and WIND databases provide comprehensive financial and market data, as well as macroeconomic, and sociodemographic information, on Chinese listed companies. These databases are extensively utilized for financial research in China and facilitate the acquisition of the necessary data for measuring population aging, digital transformation, and control variables. Accordingly, the data required for this study were sourced from these three databases. The initial sample comprised Chinese A-share listed companies on the Shanghai and Shenzhen stock exchanges from 2008 to 2020. To ensure data quality, the raw data were filtered by excluding (1) enterprises in the financial industry enterprises; (2) ST, *ST, and delisted samples during the study period; and (3) observations with missing key variables. To mitigate the impact of extreme values, all continuous variables were winsorized at the 1 % level. This yielded a final dataset of 3720 listed companies with 30,501 firm-year observations. Industries were categorized according to the 2012 China Securities Regulatory Commission industry classification table, with manufacturing classified by secondary industry and other industries by primary industry for regression analysis.

4.2. Measurement of variables

The dependent variable was corporate digital transformation (*Digital*). Text analysis was used to measure this construct and identify and capture keywords related to digital transformation in listed companies' annual reports. The frequency of these keywords served as an indicator

of the level of corporate digitalization.[3] The methodology comprised the following steps. First, a set of 73 keywords pertinent to digital transformation,[4] as constructed by Wu et al. (2021), were selected as foundational indicators for gauging corporate digital transformation. Second, annual reports from Chinese A-share listed companies from 2008 to 2020 were downloaded from the East Money website.[5] Relevant keywords associated with digital transformation were identified in these reports and tallied based on their frequency of occurrence. Third, the total count of all digital transformation-related keyword mentions within a company's annual report for a given year was aggregated, and the natural logarithm was taken as the proxy variable for the firm's digital transformation level for that year.

The independent variable was population aging (*Agingl*). This was quantified by calculating the proportion of individuals aged 65 years and older relative to the overall population in the province where each company is registered. Higher values indicated a greater degree of population aging.

Referring to Sheng et al. (2024) and Ferreira et al. (2019), this study controlled for various factors at both the firm and regional levels that could influence corporate digital transformation. The firm-level control variables included size (*Size*), return on assets (*Roa*), leverage ratio (*Lev*), cash holding level (*Cash*), revenue growth rate (*Growth*), R&D investment intensity (*RD*), intangible asset level (*Intan*), firm age (*Age*), ownership nature (*Soe*), board size (*Boardsize*), institutional ownership (*Institution*), and additional firm characteristics. Additionally, the proportion of male executives (*Male_manager*) and average age of executives (*Manager_age*) were included as entrepreneurial characteristics. At the regional level, the control variables included Internet penetration rate (*Internet_ratio*), unemployment rate (*Jobless*), economic output per capita (*GDP_per*), and regional education level (*Education*). Finally, to account for potential temporal and industry-specific effects, year and industry dummies were incorporated into the analysis. Table 1 presents a comprehensive list of the variable definitions.

4.3. Model specification

To explore the impact of population aging on corporate digital transformation, the following multiple linear regression equation is constructed[6]:

$$Digitalit = a0 + a1^{*}Aging_levelit + ai^{*}\sum controlvarit + aj^{*}Year^{*}Province + ak^{*}Industry + \varepsilon it \quad (1)$$

Table 1
Variable definitions.

Variable name	Variable definition	Variable measurement
Digital_index	Degree of digital transformation	Referring to Wu et al. (2021), the number of digital keywords in a company's annual report is used as the proxy variable of the degree of corporate digital transformation. In the analysis, logarithmic form of the number of keywords is used.
Aging	Population aging	The population over 65 years of age /the total population of the province where the enterprise is registered
Size	Enterprise size	Logarithm of year-end total assets
Lev	Leverage ratio	Total liabilities/total assets
Cash	Cash holding level	Balance of cash and cash equivalents at the end of the period/total asset
Roa	Return on assets	Net profit/total asset
Growth	Revenue growth ratio	(Revenue for current year-Revenue for previous year)/Revenue for previous year
RD	R&D investment level	R&D expenditure/total assets
Age	Enterprise age	The logarithm of the enterprise's establishment year
Hightech	High-tech enterprise	When the enterprise is a high-tech enterprise, the value is 1; otherwise, it is 0
Soe	Ownership nature	The value of state-owned enterprises is 1 and that of non-state-owned enterprises is 0.
Indep	Proportion of independent directors	Number of independent directors/ total number of directors
Gender_diversity	Executive gender diversity	Referring to Bear et al. (2010), the larger the value of Blauindex, the higher the gender heterogeneity of executives.
Manager_age	Age of executives	The logarithm of the average age of executives
Internet_ratio	Internet coverage	Internet coverage in the region of the enterprise
Jobless	unemployment rate	Registered urban unemployment rate
GDP_per	Economic level	The logarithm of the per capita GDP of the region where the enterprise is located
Education	Educational level	Proportion of college students relative to the total provincial/regional population

[3] The terminology used in annual reports reflects a company's strategic characteristics and future prospects. As digital transformation represents a significant strategy for high-quality development in the new era, related information is likely to be reflected in corporate annual reports. Therefore, characterizing the degree of transformation from the perspective of word frequency statistics related to "corporate digital transformation" in listed companies' annual reports is both feasible and scientifically valid.

[4] The keyword clusters for digital transformation encompass the following five categories:Keywords related to artificial intelligence technology: Artificial Intelligence, Business Intelligence, Image Understanding, Investment Decision Support Systems, Intelligent Data Analysis, Intelligent Robots, Machine Learning, Deep Learning, Semantic Search, Biometric Technology, Facial Recognition, Voice Recognition, Identity Verification, Autonomous Driving, Natural Language Processing. Keywords related to big data technology: Big Data, Data Mining, Text Mining, Data Visualization, Heterogeneous Data, Credit Reporting, Augmented Reality, Mixed Reality, Virtual Reality. Keywords related to cloud computing technology: Cloud Computing, Stream Computing, Graph Computing, In-memory Computing, Multi-party Security Computing, Brain-like Computing, Green Computing, Cognitive Computing, Fusion Architecture, Hundred Million Concurrent Users, Exabyte Level Storage, Internet of Things, Cyber-Physical Systems. Keywords related to blockchain technology: Blockchain, Digital Currency, Distributed Computing, Differential Privacy Technology, Smart Financial Contracts. Keywords indicating the practical application of digital technology: Mobile Internet, Industrial Internet, Mobile Connectivity, Internet Medicine, *E*-commerce, Mobile Payment, Third-party Payment, NFC Payment, Smart Energy, B2B (Business-to-Business), B2C (Business-to-Consumer), C2B (Consumer-to-Business), C2C (Consumer-to-Consumer), O2O (Online-to-Offline), Networking, Smart Wearables, Smart Agriculture, Smart Transportation, Smart Healthcare, Smart Customer Service, Smart Home, Smart Investment Advisory, Smart Culture and Tourism, Smart Environmental Protection, Smart Grid, Smart Marketing, Digital Marketing, Unmanned Retailing, Internet Finance, Digital Finance, Fintech (Financial Technology), Quantitative Finance, Open Banking.

[5] https://www.eastmoney.com/

[6] The rationale behind selecting the multiple linear regression model lies in its capacity to concurrently account for the impact of numerous independent variables on the dependent variable, thereby offering a more comprehensive explanation of the intricate relationship between population aging and corporate digital transformation. This model boasts enhanced explanatory power and predictive capability, furnishing quantifiable analytical outcomes that facilitate a profound comprehension of the interplay among diverse factors and their specific influence on corporate digital transformation processes. Employing this model enables effective control over confounding variables, thereby augmenting the precision and reliability of the research findings.

Table 2
Full sample descriptive statistic.

VarName	Obs	Mean	SD	Min	Median	Max	Skewness	Kurtosis
Digital_index	30,501	1.162	1.347	0.000	0.693	4.913	0.970	2.915
Aging	30,501	0.109	0.026	0.066	0.108	0.163	0.280	2.129
Size	30,501	22.163	1.283	19.793	21.987	26.098	0.719	3.403
Lev	30,501	0.437	0.205	0.056	0.433	0.894	0.123	2.186
Cash	30,501	0.158	0.121	0.009	0.124	0.598	1.461	5.087
Roa	30,501	0.034	0.064	−0.289	0.035	0.191	−1.804	11.005
Growth	30,501	0.165	0.412	−0.596	0.103	2.581	2.871	16.300
RD	30,501	0.017	0.019	0.000	0.012	0.093	1.627	6.188
Age	30,501	2.888	0.308	2.079	2.890	3.555	−0.325	2.820
Hightech	30,501	0.564	0.496	0.000	1.000	1.000	−0.257	1.066
Soe	30,501	0.393	0.488	0.000	0.000	1.000	0.437	1.191
Indep	30,501	0.389	0.100	0.000	0.375	0.667	−0.393	5.594
Board_diversity	30,501	0.273	0.130	0.000	0.278	0.498	−0.232	2.263
Manager_age	30,501	3.890	0.065	3.723	3.893	4.032	−0.238	2.778
Internet_ratio	30,501	0.584	0.154	0.202	0.603	0.827	−0.484	2.550
Jobless	30,501	0.031	0.007	0.013	0.031	0.043	−0.537	3.063
GDP_per	30,501	2.013	0.433	0.957	2.037	2.862	−0.211	2.564
Education	30,501	0.029	0.009	0.014	0.026	0.056	1.433	4.581

The dependent variable, $Digital_{it}$, represents the level of digital transformation of firm i in year t. The independent variable, $Aging_level_{it}$, represents the level of population aging in the province where the firm is registered in year t. This study focused on regression coefficient a_1, which indicates the extent to which the average change in corporate digital transformation corresponds to each additional unit of population aging. If a_1 is greater than zero, it implies that population aging promotes corporate digital transformation, validating H1a. Otherwise, it suggests that population aging hinders corporate digital transformation, validating H1b. $controlvar_{it}$ represents a series of control variables. Finally, the regression model controls for time–province fixed effects. *Year*×*Province*, and industry fixed effects, *Industry*, with ε_{it} representing the error term. To address potential clustering features in the sample data, the standard errors of the regression coefficients were adjusted at the firm level using clustering (*cluster*).

5. Empirical results

5.1. Descriptive statistics

Table 2 presents descriptive statistics for the main variables. The mean value of corporate digital transformation (*Digital*) is 1.162, with a maximum value of 4.913, indicating significant variation in the level and pace of digital transformation among Chinese enterprises. The mean value of population aging (*Aging*) is 0.109, with a standard deviation of 0.026, suggesting a high degree of aging in China, with substantial variation across provinces. Descriptive statistics for the other control variables are also presented in Table 2.

5.2. Regression results of population aging and corporate digital transformation

To examine the impact of population aging on corporate digital transformation, we employed the multiple linear regression model described above. Table 3 presents the results. In column (1), controlling only for industry, time, and individual fixed effects, the coefficient of population aging is 50.411, which is significant at the 1 % level. In column (2), after adding the firm and entrepreneur characteristic variables to column (1), the coefficient for population aging is 42.372 and significant at the 1 % level. In column (3), with the addition of regional control variables to column (2), the coefficient for population aging is 25.636 and significant at the 1 % level.

The regression coefficients for population aging in Table 3 are significantly positive, indicating that population aging promotes corporate digital transformation. Notably, after controlling for firm- and region-level factors, the coefficient of population aging is the smallest. This suggests that the influence of firm- and region-level factors on corporate digital transformation cannot be ignored; otherwise, the impact of population aging on corporate digital transformation would be overestimated. Overall, these findings support H1a.

5.3. Robustness tests

First, we replaced the measure of corporate digital transformation. To depict corporate digital transformation more comprehensively, following He and Hongxia (2019), we reconstructed a keyword group describing corporate digital transformation based on policy documents related to the digital economy or digital technology issued by the Chinese government, creating an alternative indicator (*PPL*)[7] for corporate digital transformation, and re-ran the regression tests. As shown in column (1) of Table 4, the coefficient of population aging and corporate digital transformation is 32.622 and significant at the 1 % level.

Second, we replaced the measure of population aging. Using the older adult support ratio (*Old_support*) as an alternative indicator of population aging, which represents the number of older adults per 100 working-age individuals, we examined the social consequences of population aging from an economic perspective. The regression results are presented in column (2) of Table 4, with the coefficient of population aging and corporate digital transformation being 16.276and significant at the 1 % level.

Third, we replaced the sample based on place of registration with one based on place of operation. Firms may have offices in their place of registration but conduct specific recruitment and production operations in their actual place of business, where the degree of aging might differ from that of the registered location. To better ensure the robustness of the previous conclusions, we ran the regression again using the population aging level of a firm's actual place of business instead of its province of registration. The results are shown in column (3) of Table 4, with the coefficient for population aging and corporate digital

[7] The specific approach involved three steps. First, we searched the website of the Ministry of Industry and Information Technology of China (www.miit.gov.cn) to download and manually filter 188 government documents related to the digital economy or digital technology published between 2008 and 2020. Second, we used Python for word segmentation and manual identification to obtain a total of 134 keywords related to digital transformation across 11 subdimensions, including digitalization, intelligentization, application of digital technologies, big data, cloud computing, artificial intelligence, machine learning, blockchain, Internet of Things, virtual reality and augmented reality, and other digital technologies. Third, we identified and extracted keywords from corporate annual reports and calculated their frequency as a proxy indicator (*PPL*) for corporate digital transformation.

Table 3
Regression results of population aging and digital transformation.

	(1)	(2)	(3)
Aging	50.411***	42.372***	25.636***
	(5.539)	(4.772)	(3.148)
Size		0.169***	0.169***
		(13.223)	(13.223)
Lev		−0.067	−0.067
		(−0.819)	(−0.819)
Cash		0.611***	0.611***
		(5.754)	(5.754)
Roa		−0.613***	−0.613***
		(−3.624)	(−3.624)
Growth		0.078***	0.078***
		(4.534)	(4.534)
RD		12.237***	12.237***
		(12.493)	(12.493)
Age		0.022	0.022
		(0.455)	(0.455)
Hightech		0.090*	0.090*
		(1.695)	(1.695)
Soe		−0.151***	−0.151***
		(−4.769)	(−4.769)
Indep		0.196**	0.196**
		(1.984)	(1.984)
Board_diversity		0.355***	0.355***
		(3.486)	(3.486)
Manager_age		−1.367***	−1.367***
		(−6.142)	(−6.142)
Internet_ratio			8.811**
			(2.458)
Jobless			−145.371***
			(−4.280)
GDP_per			2.512***
			(2.939)
Education			12.554
			(0.638)
_cons	−6.514***	−3.921***	−6.726**
	(−5.346)	(−2.799)	(−2.258)
Year×Province	Yes	Yes	Yes
Industry	Yes	Yes	Yes
N	30,501	30,501	30,501
Adj. R^2	0.409	0.447	0.447

Note: *, * *, * * * mean significant at 10 %, 5 %, 1 % level respectively.

transformation being 36.022 and significant at the 1 % level.

Fourth, we excluded samples in which the provinces of operation and registration were inconsistent. As mentioned above, significant differences in the aging process can be observed among Chinese provinces. Given that the degree of population aging in the province of registration may differ from that in the actual place of business, leading to potential errors in the regression results, we further excluded samples with inconsistent provinces of operation and registration and ran the regression again. The results are in column (4) of Table 4 show that the coefficient for population aging and corporate digital transformation is 34.576 and significant at the 1 % level.

Finally, we excluded samples from provinces with a gradual aging trend and ran the regression again. The results are shown in column (5) of Table 4, with the coefficient for population aging and corporate digital transformation being 34.913 and significant at the 1 % level.

Similar to the results in Table 3, all the regression coefficients for population aging in Table 4 are significantly positive. These findings provide further support for H1a, indicating that population aging promotes corporate digital transformation.

5.4. Endogeneity control

To address potential endogeneity concerns, such as the bidirectional causal relationship between population aging and corporate digital transformation, as well as omitted variable bias in empirical studies, we employed several econometric strategies: variable lagging, instrumental variable (IV) approaches, and propensity score matching with the difference-in-differences (PSM-DID) model.

First, variable lagging was implemented. Recognizing that the impact of population aging on corporate digital transformation might exhibit a temporal delay, we introduced a one-period lag for the core explanatory variable to account for practical transmission delays and technically mitigate reverse causality interference. The results are presented in column (1) of Table 5. The coefficient for population aging is 27.602 and significant at the 1 % level, which aligns with the baseline regression results in Table 3.

Second, we adopted the IV method. Following Cai and Stoyanov (2016), we utilized historical variables as instruments for population aging, employing early birth rates from 1978 to 1990 as instruments for population aging from 2008 to 2020 to alleviate endogeneity issues. The regression findings are reported in column (2) of Table 5. The coefficient for population aging and digital transformation is 25.663 and significant at the 1 % level, which is consistent with the baseline regression results in Table 3.

Third, we implemented the PSM-DID approach. As mentioned above, each province has a different level of population aging, and the level of population aging faced by companies undergoing interprovincial relocation varies. We constructed a DID model based on events of interprovincial relocation of company registration.

First, we conducted the group assignment. We initially identified companies that underwent inter-provincial relocation during the sample period, determined the timing of these relocations, and excluded companies that had undergone multiple relocations. We then defined companies that relocated from provinces with lower levels of aging to those with higher levels as the treatment group (treat = 1) and those that did not undergo any relocations during the period as the control group (treat = 0). Finally, we set the value of *post* to 1 for the year of interprovincial relocation and subsequent years and 0 otherwise.

Second, we matched the sample. The matching variables included company size, debt-to-asset ratio, profitability, years since establishment, and cash holding levels. Using a 1:2 nearest neighbor matching method annually, we matched two control group companies with similar characteristics to each treated company. This resulted in 26 treated and 49 control companies (with some overlapping matches), which showed no significant differences in the covariates between groups. To avoid interference from other factors over an extended window, we focused on digitalization level fluctuations within three years before and after companies relocated to provinces with higher levels of aging. Here, t = 0 represents the year of the inter-provincial relocation.

Third, we applied the DID model. The robustness regression results of PSM-DID are shown in Table 6. After controlling for sample selection bias and company characteristic differences, the results are consistent with the baseline regression (Table 3). These results indicate that population aging promotes corporate digital transformation, thus supporting H1a.

6. Heterogeneity tests

The above analysis verifies that population aging promotes corporate digital transformation. However, different industries have varying labor demands, and the impact of population aging varies significantly. Additionally, given China's vast territory, substantial regional differences can be found in the aging process, leading to varying degrees of aging-related challenges facing enterprises in different areas. Therefore, we examined the heterogeneity of the impact of population aging on corporate digital transformation from both industrial and regional perspectives.

6.1. Industry heterogeneity test

Given the substantial variations in the repercussions of population aging across distinct industries, its influence on corporate digital transformation could plausibly exhibit industry-specific heterogeneity.

Table 4
Robustness test.

	(1)	(2)	(3)	(4)	(5)
Aging	32.622***		36.022***	34.576***	34.913***
	(3.407)		(4.572)	(4.322)	(4.763)
Old_support		16.276***			
		(3.148)			
Size	0.139***	0.169***	0.166***	0.167***	0.167***
	(9.523)	(13.223)	(12.873)	(12.620)	(11.234)
Lev	−0.019	−0.067	−0.071	−0.056	−0.146
	(−0.207)	(−0.819)	(−0.875)	(−0.672)	(−1.616)
Cash	0.442***	0.611***	0.600***	0.606***	0.632***
	(3.804)	(5.754)	(5.654)	(5.599)	(5.226)
Roa	−0.225	−0.613***	−0.620***	−0.674***	−0.611***
	(−1.157)	(−3.624)	(−3.669)	(−3.850)	(−3.095)
Growth	0.092***	0.078***	0.074***	0.081***	0.072***
	(4.601)	(4.534)	(4.247)	(4.428)	(3.561)
RD	13.614***	12.237***	12.310***	12.462***	11.273***
	(12.891)	(12.493)	(12.517)	(12.600)	(8.953)
Age	−0.058	0.022	0.022	0.022	−0.049
	(−1.084)	(0.455)	(0.451)	(0.451)	(−0.830)
Hightech	0.168***	0.090*	0.091*	0.084	0.074
	(2.666)	(1.695)	(1.713)	(1.546)	(1.239)
Soe	−0.114***	−0.151***	−0.143***	−0.148***	−0.109***
	(−3.343)	(−4.769)	(−4.461)	(−4.570)	(−3.037)
Indep	0.181	0.196**	0.205**	0.223**	0.085
	(1.632)	(1.984)	(2.050)	(2.275)	(0.794)
Board_diversity	0.431***	0.355***	0.362***	0.366***	0.314***
	(3.951)	(3.486)	(3.540)	(3.511)	(2.711)
Manager_age	−0.494**	−1.367***	−1.330***	−1.239***	−1.288***
	(−1.977)	(−6.142)	(−5.984)	(−5.537)	(−4.978)
Internet_ratio	13.624***	2.486	0.076**	8.212**	8.447***
	(3.380)	(0.498)	(2.107)	(2.201)	(3.587)
Jobless	−197.172***	−117.465***	−1.824***	−177.681***	−164.162***
	(−5.386)	(−4.157)	(−5.390)	(−5.102)	(−4.783)
GDP_per	4.997***	0.601	2.338***	2.719***	2.369***
	(5.389)	(0.520)	(3.216)	(3.102)	(4.170)
Education	−35.542	30.538	36.986**	36.390*	34.000
	(−1.543)	(1.245)	(1.990)	(1.921)	(1.463)
_cons	−14.348***	−0.238	−27.825***	−8.115***	−7.629**
	(−4.469)	(−0.058)	(−3.100)	(−2.639)	(−2.411)
Year×Province	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
N	21,787	30,501	30,493	29,082	22,257
Adj. R^2	0.517	0.447	0.447	0.451	0.407

Note: *, * *, * * * mean significant at 10 %, 5 %, 1 % level respectively.

To elucidate this, we stratified companies into three categories–technology-, capital-, and labor-intensive. The regression analyses were conducted separately for each category, and the results are presented in Table 7. The findings indicate that the effect of population aging on digital corporate transformation varies across industries.

In technology- and capital-intensive industries (Table 7, columns 1 and 2), the coefficients of population aging are positive but not statistically significant. In contrast, in labor-intensive industries (Table 7, column 3), the coefficient for population aging is 35.544 and significant at the 1 % level. These findings indicate that the impact of population aging on corporate digital transformation is stronger in labor-intensive enterprises. This could be attributable to the higher demand for young labor in labor-intensive enterprises, whereas population aging leads to a decrease in the proportion of young people in the labor market and an increase in labor costs. Labor-intensive industries face greater effects and challenges, and thus have a stronger motivation for corporate digital transformation. Capital- and technology-intensive industries typically already possess a strong technological foundation and adaptability, allowing them to adjust their production processes and management models more flexibly to cope with the challenges and pressures caused by population aging. Consequently, their willingness to undergo digital transformation is lower. This reveals that the impact of population aging on corporate digital transformation exhibits some industry heterogeneity, further validating H1a from another perspective.

6.2. Regional heterogeneity

China's landscape is characterized by marked disparities in the extent and tempo of population aging across its geographic expanse, implying a differential exposure of regional enterprises to the ramifications thereof. This context necessitates an examination of potential regional heterogeneity in the effect of population aging on corporate digital transformation. Accordingly, our sample was segmented into four geographical zones: eastern, central, western, and northeastern China.[8] Individual regression assessments were conducted for each region. The ensuing data, spanning columns (4)–(7) of Table 7, illustrate the nuanced regional effects. The northeastern region (column 4) manifests a coefficient of 14.582, which is not statistically significant. The western and central regions (columns 5 and 6) have coefficients of 12.392 and −25.286, respectively. In contrast, the eastern region (column 7) has a coefficient of 13.702 that is significant at the 1 % level, highlighting

[8] The Northeast region includes Liaoning, Jilin, and Heilongjiang (three provincial-level administrative regions); the Eastern region comprises Beijing, Tianjin, Hebei, Shandong, Jiangsu, Shanghai, Zhejiang, Fujian, Guangdong, and Hainan (ten provincial-level administrative regions); the Central region encompasses Shanxi, Henan, Anhui, Hubei, Jiangxi, and Hunan (six provincial-level administrative regions); and the Western region consists of Chongqing, Sichuan, Shaanxi, Yunnan, Guizhou, Guangxi, Gansu, Qinghai, Ningxia, Tibet, Xinjiang, and Inner Mongolia (twelve provincial-level administrative regions).

Table 5
Robustness result: controlling the endogenous.

	(1)	(2)
	Lag Processing	IV
Aging	27.602***	25.663***
	(3.273)	(3.177)
Size	0.161***	0.173***
	(12.023)	(13.324)
Lev	−0.033	−0.077
	(−0.381)	(−0.932)
Cash	0.684***	0.611***
	(6.208)	(5.671)
Roa	−0.517***	−0.632***
	(−2.706)	(−3.711)
Growth	0.092***	0.086***
	(5.047)	(4.770)
RD	12.376***	12.460***
	(12.288)	(12.652)
Age	0.021	0.025
	(0.394)	(0.499)
Hightech	0.097*	0.093*
	(1.677)	(1.720)
Soe	−0.167***	−0.158***
	(−4.947)	(−4.897)
Indep	0.292***	0.194**
	(2.812)	(1.988)
Board_diversity	0.377***	0.364***
	(3.506)	(3.514)
Manager_age	−1.468***	−1.378***
	(−6.285)	(−6.108)
Internet_ratio	3.333	8.726**
	(0.918)	(2.453)
Jobless	−116.849***	−145.581***
	(−3.345)	(−4.312)
GDP_per	1.663*	2.492***
	(1.899)	(2.935)
Education	−2.434	13.053
	(−0.120)	(0.669)
_cons	−1.923	−6.696**
	(−0.640)	(−2.261)
Year×Province	Yes	Yes
Industry	Yes	Yes
N	29,736	29,037
Adj. R^2	0.436	0.442

Note: *, * *, * * * mean significant at 10 %, 5 %, 1 % level respectively.

Table 6
Robustness result: propensity score matching with differences-in-difference.

	(1)	(2)	(3)
	[−3,3]	[−2,2]	[−1,1]
DID	0.400	0.157	−0.114
	(0.829)	(0.323)	(−0.283)
treat	−0.219	−0.026	0.131
	(−0.494)	(−0.051)	(0.329)
post	−0.071	0.222	0.911**
	(−0.355)	(0.923)	(2.524)
Size	0.131	0.210**	0.295***
	(1.601)	(2.338)	(3.464)
Lev	1.219	0.712	−0.055
	(1.642)	(1.054)	(−0.107)
Cash	1.635*	1.929	1.083
	(1.916)	(1.657)	(0.736)
Roa	2.526*	3.122**	2.752*
	(1.682)	(2.134)	(1.924)
Growth	0.080	−0.133	−0.308*
	(0.680)	(−1.188)	(−1.790)
RD	28.964***	22.676***	13.971**
	(5.447)	(3.562)	(2.018)
Age	0.760	1.045*	1.159**
	(1.246)	(1.789)	(2.567)
Hightech	0.029	0.109	0.429
	(0.078)	(0.292)	(1.160)
Soe	−0.758**	−0.644**	−0.304
	(−2.601)	(−2.163)	(−1.316)
Indep	2.039**	1.852*	0.896
	(2.523)	(2.000)	(1.358)
Board_diversity	1.347	0.511	0.525
	(1.394)	(0.468)	(0.671)
Manager_age	−1.801	−2.849	−5.214**
	(−0.897)	(−1.271)	(−2.334)
Internet_ratio	4.532	8.422	16.725
	(0.577)	(1.102)	(1.335)
Jobless	−0.711	−0.969*	−0.815
	(−1.365)	(−1.805)	(−0.863)
GDP_per	−2.766	−2.898	−3.931
	(−1.034)	(−1.011)	(−0.886)
Education	−47.928	2.721	65.444
	(−0.698)	(0.036)	(0.803)
_cons	8.461	8.014	10.307
	(0.866)	(0.713)	(0.978)
Year×Province	Yes	Yes	Yes
Industry	Yes	Yes	Yes
N	1032	810	531
Adj. R^2	0.775	0.792	0.900

Note: *, * *, * * * mean significant at 10 %, 5 %, 1 % level respectively.

discernible regional specificity in how population aging influences corporate digital transformation. This could be attributable to the relatively higher degree of aging in the eastern region than in other areas. Consequently, enterprises in the eastern region face greater effects and challenges from an aging population, creating a stronger incentive for them to undergo digital transformation. These findings suggest that the influence of population aging on corporate digital transformation exhibits regional heterogeneity, further validating H1a.

7. Conclusions, implications, and limitations

7.1. Conclusions

Since the 20th century, with declining fertility rates and increasing life expectancies, the global population structure has undergone an unprecedented transformation, with most countries entering a stage of demographic aging. Understanding how population aging affects corporate digital transformation is crucial for addressing the challenges related to aging populations and innovating digital technology strategies.

First, this study systematically analyzed the theoretical implications of population aging on corporate digital transformation and proposed two opposing hypotheses based on the potential dual effects of aging: population aging promotes corporate digital transformation (H1a) and hinders such transformation (H1b). Subsequently, using a multiple linear regression model and panel data from 3720 listed companies in China's Shanghai and Shenzhen stock markets from 2008 to 2020, we empirically investigated the impact of population aging on corporate digital transformation and its heterogeneity across industries and regions.

The results are as follows. First, a significant positive correlation exists between population aging and corporate digital transformation, confirming H1a. This conclusion remains robust after considering endogeneity issues, changing the variable measurement methods, and altering the test sample, among other robustness checks. Second, the effects of population aging on corporate digital transformation vary significantly across industries and regions. Specifically, compared to capital- and technology-intensive firms that are less affected by aging, labor-intensive firms experience a greater promotional effect on digital transformation due to population aging. Moreover, this promotional effect is more pronounced in enterprises located in China's eastern regions. These findings regarding heterogeneity reinforce the positive correlation between population aging and corporate digital transformation, further validating H1a.

7.2. Implications

This study also makes some theoretical contributions. First, this study provides an initial examination of the influence of population

Table 7
Heterogeneity test: industry and regional.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Aging	18.991	19.538	35.544**	14.582	12.392	−25.286	13.702**
	(1.431)	(1.433)	(3.085)	(0.752)	(1.534)	(−0.850)	(3.638)
Size	0.178***	0.155**	0.181***	0.117*	0.143**	0.132**	0.198***
	(7.620)	(6.700)	(9.408)	(2.523)	(4.859)	(4.166)	(12.214)
Lev	−0.093	−0.136	−0.069	−0.354	0.036	−0.371*	−0.036
	(−0.660)	(−1.009)	(−0.558)	(−1.178)	(0.197)	(−1.838)	(−0.343)
Cash	0.018	0.389*	0.504***	0.394	0.175	0.275	0.719***
	(0.098)	(1.811)	(3.161)	(0.951)	(0.771)	(0.919)	(5.562)
Roa	−0.424	−0.877***	−0.223	0.157	−0.060	−0.913**	−0.798***
	(−1.359)	(−2.906)	(−0.879)	(0.248)	(−0.147)	(−2.139)	(−3.779)
Growth	0.113***	0.134***	0.049**	0.112*	0.069*	0.034	0.083***
	(3.213)	(3.316)	(2.030)	(1.765)	(1.807)	(0.681)	(3.817)
RD	5.581***	0.083	6.361***	3.308	12.580***	8.756***	13.177***
	(4.247)	(0.046)	(3.461)	(0.608)	(4.093)	(3.620)	(11.432)
Age	0.123	−0.032	−0.106	−0.074	0.012	−0.272**	0.097
	(1.383)	(−0.311)	(−1.432)	(−0.385)	(0.092)	(−1.972)	(1.616)
Hightech	0.018	−0.001	0.058	−0.318	0.018	0.309***	0.063
	(0.187)	(−0.017)	(0.169)	(−0.992)	(0.146)	(3.040)	(0.929)
Soe	−0.175***	−0.234***	−0.132***	−0.174	−0.063	−0.084	−0.183***
	(−2.907)	(−4.330)	(−2.957)	(−1.294)	(−0.835)	(−1.157)	(−4.369)
Indep	−0.014	0.361*	−0.008	−0.404	0.270	0.145	0.239**
	(−0.088)	(1.855)	(−0.050)	(−1.090)	(1.011)	(0.624)	(1.979)
Board_diversity	0.050	0.195	0.422***	0.475	−0.074	0.005	0.454***
	(0.296)	(1.040)	(2.780)	(1.060)	(−0.288)	(0.023)	(3.559)
Manager_age	−0.676*	−1.110***	−1.463***	−0.139	−0.436	−2.213***	−1.516***
	(−1.759)	(−2.677)	(−4.424)	(−0.131)	(−0.724)	(−3.763)	(−5.768)
Internet_ratio	13.892**	13.009*	4.182	5.693***	4.002***	−3.524	−11.697***
	(2.027)	(1.751)	(0.724)	(2.771)	(2.785)	(−0.682)	(−2.586)
Jobless	−158.435**	−145.661	−144.920***	38.109	−27.722	19.678	173.348***
	(−2.397)	(−1.406)	(−3.019)	(1.310)	(−0.855)	(0.297)	(3.235)
GDP_per	4.538***	3.275	1.120	−3.964**	−1.790***	3.660	1.333***
	(2.681)	(1.559)	(0.947)	(−2.148)	(−2.653)	(0.779)	(4.609)
Education	−34.038	23.834	24.835	38.122	16.701	19.204	−187.068***
	(−1.153)	(0.573)	(0.916)	(1.566)	(0.999)	(0.276)	(−4.009)
_cons	−13.985**	−10.686	−2.353	−1.818	−0.600	5.745*	4.806**
	(−2.314)	(−1.520)	(−0.529)	(−0.406)	(−0.230)	(1.945)	(2.130)
Year×Province	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	8373	5713	11,485	1525	4289	4349	20,338
Adj. R^2	0.612	0.228	0.445	0.419	0.419	0.433	0.455

Note: *, * *, * * * mean significant at 10 %, 5 %, 1 % level respectively.

aging on corporate digital transformation. This complements the works of Acemoglu and Restrepo (2022) and Ang and Madsen (2015) by affirming the stimulatory role of population aging in fostering technological innovation. However, in contrast to these studies, which primarily explored automation technology (Acemoglu and Restrepo, 2022) or compared the innovativeness of older versus younger workers with higher education (Ang and Madsen, 2015), our study concentrates on corporate digital transformation. Digital transformation transcends traditional automation and information technologies, engendering shifts not only within organizational structures but also across business models that are inadequately addressed by extant theoretical frameworks (Markus and Rowe, 2018). Building on Coccia's extensive exploration of the origins of disruptive innovation (Coccia, 2010, 2014a, 2014b, 2014c, 2015a, 2015b, 2017a, 2017b), this study broadens the analytical lens to encompass the impact of population aging on corporate digital transformation in emerging markets. This empirical investigation fills a notable gap in the literature on the effects of an aging population on technological innovation within such contexts. Furthermore, it contributes to the discourse on the economic implications of population aging by highlighting positive outcomes at the micro-enterprise level, an underexplored area identified by Acemoglu and Restrepo (2022) as a future research avenue. Heterogeneity tests further illuminate the nuanced effects, offering a foundation for subsequent inquiries into the diverse facets of corporate decision-making influenced by demographic shifts.

Based on this study's findings, we propose the following practical implications from both governmental and enterprise perspectives. From the governmental perspective, bolstering top-tier design guidance and policy orientation is imperative. First, when developing policies to foster corporate digital transformation, governments must thoroughly consider industry-specific attributes and regional disparities. By offering tax incentives, R&D subsidies, and low-interest loans to alleviate the financial burden of digital transformation on enterprises and tailoring these incentive policies to reflect each region's unique demographic structure, resource endowments, cultural traits, and other factors, a digital economy tailored to local geographic conditions can be promoted. Second, enhancing investment in Internet infrastructure and related technologies, particularly in remote areas, ensures ubiquitous access to network services, thereby promoting the equitable development of the digital economy across diverse regions. Third, intensifying regional educational initiatives to holistically elevate workforce human capital will supply the ample talent necessary for local enterprises to engage in digital transformation. Finally, organizing specialized training programs encompassing basic computer literacy, smart device utilization, and cybersecurity awareness can aid older adults in enhancing their digital proficiency, thus enabling them to seamlessly integrate into a digital lifestyle.

From the enterprise perspective of enterprises, first an escalation in the recognition of the significance of digital transformation is needed. With population aging leading to labor shortages and escalating labor costs becoming increasingly prevalent, companies must prioritize digital strategies and initiate digital transformations at strategically opportune times to adapt to environmental shifts and secure a sustainable competitive edge. Second, exploring the distinct needs and usage

patterns of older adults will enable the design of more user-centric interfaces and functionalities, thereby better catering to the heterogeneous requirements of older users. Finally, enterprises should prioritize human capital accumulation as a buffer against the detrimental impact of population aging on human resources. Initiating efforts to attract and nurture high-caliber human capital, such as actively recruiting digital experts and reinforcing employees' digital technology knowledge and skills through training, will amplify the competitive advantage stemming from an enterprise's human capital reserves, thereby laying a solid foundation for the smooth progression of digital transformation.

7.3. Limitations

A primary limitation of this study is that the data source is restricted to China, a nation that is experiencing rapid population aging and has unique sociocultural and economic attributes that may constrain the generalizability of the findings to other nations. Future studies that incorporate cross-national datasets should be encouraged to enhance the external validity. Another limitation is the focus on listed companies, which typically enjoy more favorable environmental conditions and resource endowments than their non-listed counterparts. The latter often struggle with the capacity and resources necessary for digital transformation, raising questions regarding whether similar aging-induced transformation dynamics apply to non-listed companies. Including non-listed firms in future research could refine our understanding and broaden the applicability of our results.

CRediT authorship contribution statement

Ping Zhang: Writing – review & editing, Writing – original draft, Data curation. **Huilin Yang:** Data curation. **Chen Chen:** Writing – review & editing, Writing – original draft. **Tao Wang:** Writing – review & editing, Writing – original draft. **Ximeng Jia:** Methodology.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Data availability

Data will be made available on request.

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